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IEEE Communications Society

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Keynote Speakers

Bruno Clerckx (Imperial College London, UK)
Davide Dardari (University of Bologna, Italy)

Important Dates

Paper Submission: 17 Jan. 2027
Acceptance Notification: 7 Mar. 2027
Camera-Ready Due: 21 Mar. 2027

Scope and Topics of Interest

Digital circuits underpin almost every modern communication system, but their high power consumption and limited processing speed become critical as antenna arrays scale toward massive and gigantic multiple-input multiple-output (MIMO) systems in 6G and beyond. Each antenna requires a dedicated radio-frequency chain with power-hungry data converters, while large-scale digital signal processing has a complexity that scales cubically with the number of antennas.

Analog computing overcomes these two bottlenecks by manipulating signals directly in the analog domain, which enables energy-efficient and massively parallelized computations. It can be realized through metamaterials and metasurfaces, microwave networks, resistive memory arrays, and optical systems, which bypass digital conversion and realize linear transforms, such as beamforming and the discrete Fourier transform, directly in the wave domain. While the main ICC 2027 symposia treat computation as a digital-domain task performed after the signal has been captured and converted, **this workshop moves the computation itself into the analog and wave domain**, and brings the metasurface, microwave-network, photonic, and device communities together under the umbrella of analog computing.

Topics of interest include, but are not limited to:

- Analytical foundations, information-theoretic limits, and unified modeling of analog computing for communications, sensing, and imaging.
- Novel analog computing architectures utilizing metasurfaces, microwave networks, resistive memory arrays, and optical systems.
- Design, training, and deployment of physical/analog neural networks.
- Programmable over-the-air computing and analog-domain nonlinear function computation.
- Analog-native physical layer architectures, multiple access, and interference management for 6G and beyond.
- Analog computing for ultra-reliable low-latency communications, integrated sensing and communications, and non-terrestrial networks.
- Cross-layer design, optimization, and computation-aware resource allocation for future beamforming and massive MIMO.
- AI-empowered design of analog computing-aided networks, and analog computing-empowered AI processing.
- Performance analysis and mitigation of hardware non-idealities, such as discrete-value and lossy reconfigurable components, mutual coupling, and insertion losses.
- Hardware implementations and testbed experimentation spanning RF, mmWave, THz, and optical frequencies.

The workshop features two keynotes given by leading researchers in this field. Submitted papers should follow the IEEE ICC paper submission guidelines. Papers should be submitted for review through EDAS. Accepted papers will be published in IEEE Xplore.